

CASE 23

X-LINKED HYPOHIDROTIC ECTODERMAL DYSPLASIA AND IMMUNODEFICIENCY

Immunodeficiency due to a defective component in an intracellular signaling pathway required for both innate and adaptive immunity.

Intracellular signaling molecules are vital to the normal maturation of B and T lymphocytes. In Case 1 we saw that X-linked agammaglobulinemia results from a failure of normal B-cell maturation because of mutations in a tyrosine kinase (Bruton's tyrosine kinase, Btk) that is vital to normal calcium movements in pre-B cells. In its absence, pre-B cells fail to progress to mature B cells and thus never become immunoglobulin-secreting plasma cells. Other signaling molecules are involved in the transmission of co-stimulatory signals to lymphocytes. In Case 2 we saw that X-linked hyper IgM syndrome results from the failure of B cells to undergo class-switch recombination and somatic hypermutation because they do not receive a signal from helper T cells, owing to a failure of these cells to express CD40 ligand.

Although some signaling molecules, such as CD40 ligand, are peculiar to lymphocytes, others are found in many cell types. A crucial step in the activation of many different cells is the induction of the transcription of specific genes by transcription factors of the NF κ B family. When not required, NF κ B and other family members reside in the cytoplasm, where they are maintained in an inactive state in a complex with their inhibitor protein, I κ B. When the cell is activated by the appropriate external signal, a cascade of protein kinase activity is induced in the

TOPICS BEARING ON THIS CASE:

Macrophage
activation

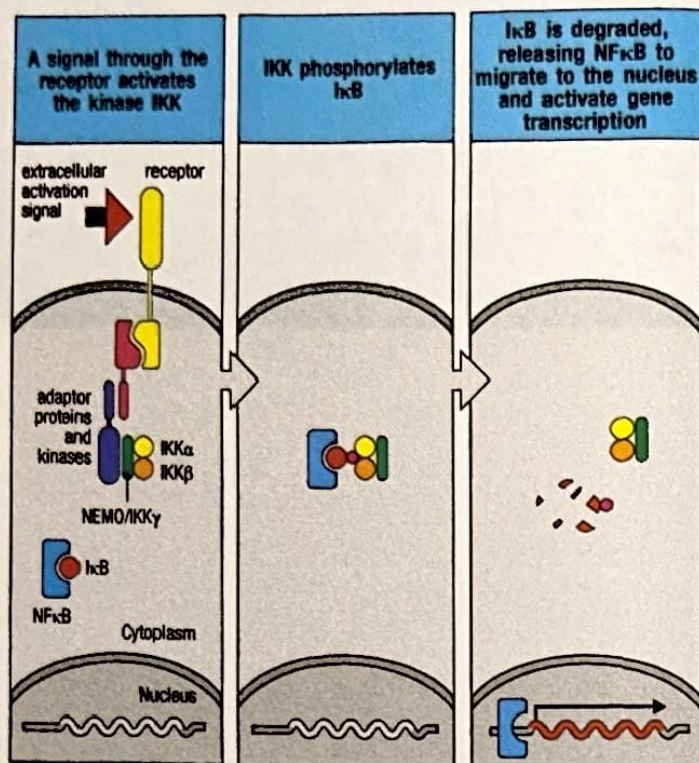
Isotype switching

Activation of NF κ B

T-cell help in
antibody response

CD40 ligand and
isotype switching

Fig. 23.1 The NF κ B activation pathway. Various receptors can activate a series of intracellular adaptor molecules and kinases that ultimately induce the formation of the IKK complex (consisting of IKK α , IKK β , and NEMO/IKK γ). The IKK complex then phosphorylates the inhibitory protein I κ B, resulting in its dissociation from cytoplasmic NF κ B and its ultimate ubiquitination and degradation. Once free from the inhibitory influence of I κ B, NF κ B translocates to the nucleus, where it induces the transcription of specific genes.



cytoplasm. This causes the assembly and activation of the I κ B kinase (IKK) complex (Fig. 23.1), which consists of three components—the kinases IKK α and IKK β along with another protein, the NF κ B essential modifier (NEMO), also known as IKK γ . The active IKK complex phosphorylates I κ B, which targets I κ B for degradation by the ubiquitin–proteasome system, thus freeing NF κ B from its inhibitor. A nuclear localization signal is uncovered on NF κ B that enables the transcription factor to move into the nucleus, where it binds to particular nucleotide sequences in the promoters of selected genes and initiates their transcription.

NF κ B is activated by the ligation of a variety of receptors, including many with fundamental roles in development, inflammation, and the generation of immune responses. For example, the binding of the pro-inflammatory cytokines tumor necrosis factor- α (TNF- α) or interleukin 1 (IL-1) to their receptors results in NF κ B activity. The Toll-like receptors (TLRs) on macrophages, which are key components of the innate immune system and recognize microbial components and other danger signals, also work through NF κ B activation. And NF κ B is activated in B cells after stimulation via CD40.

Functional NF κ B is measured in the laboratory by an electrophoretic mobility-shift assay (EMSA) that detects its presence in the nucleus. Cell nuclei are isolated by ultracentrifugation, and their NF κ B content is measured by assessing the capacity of the nuclear extract to bind to a synthetic radiolabeled oligonucleotide probe that contains NF κ B binding sites, and thus to retard the probes migration in gel electrophoresis (Fig 23.2).

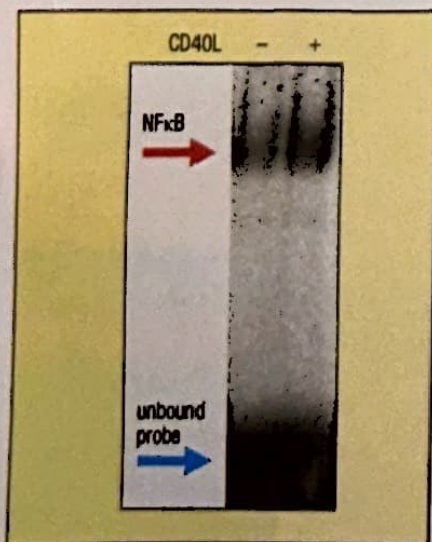


Fig. 23.2 Electrophoretic mobility-shift assay (EMSA). Nuclei were isolated from normal B cells by ultracentrifugation, and nuclear extracts were mixed with a 32 P-radiolabeled DNA probe that binds to NF κ B. Free probe, not associated with NF κ B, can be seen at the bottom of the gel, whereas probe bound to NF κ B is retarded in the gel as a result of the higher molecular weight of the protein–probe complex. Although unstimulated B cells contain a baseline level of NF κ B in their nuclei (as seen in the left lane), ligation of CD40 by CD40 ligand (CD40L) results in a higher level of NF κ B in the nucleus (as evidenced by the darker band in the right lane).

The case of Robert Teixeira: severe bacterial infections and pointed teeth.

Robert was born after an uneventful pregnancy. He received his normal immunizations at 2, 4, and 6 months, including a pneumococcal vaccine that contained seven different serotypes of *Streptococcus pneumoniae*, without any ill effects. At 9 months old he developed recurrent fever associated with nasal discharge, which was diagnosed as a viral infection of the upper respiratory tract. He began to show less interest in his bottle, cried frequently, and became increasingly difficult to console.

Three weeks after the fevers began, he had a seizure and was taken to the local emergency department. The seizure was controlled with an intravenous anticonvulsive medicine and the physician noted that his skin felt warm and had a reticular (lacy) pattern of blood vessels. Robert's neck was rigid and he screamed when his head was moved. A complete blood count revealed an abnormally high white blood cell count of $48,200 \mu\text{l}^{-1}$, 90% of which were neutrophils (normal $5000 \mu\text{l}^{-1}$). Spinal fluid was sampled by lumbar puncture and was free-flowing and cloudy. Microscopic and chemical analysis revealed 12 red blood cells per high-power field (normal 0); 11,382 white blood cells per high-power field (normal <6), of which 85% were neutrophils; a protein concentration of 410 mg dl^{-1} (elevated compared with normal); and a glucose concentration of 39 mg dl^{-1} (low compared with normal).

A Gram stain of the spinal fluid showed paired Gram-positive cocci. Robert was treated with ceftriaxone (an extended-spectrum cephalosporin antibiotic) and admitted to intensive care. He responded rapidly to the antibiotic and improved daily. A culture of his spinal fluid grew *S. pneumoniae*, which expressed the p14 cell-wall polysaccharide. An immunology consultation was requested because children receiving the seven-valent pneumococcal vaccine should be protected against strains of *S. pneumoniae* carrying this polysaccharide.

Laboratory analysis of Robert's serum immunoglobulin showed 170 mg dl^{-1} IgG (normal $400\text{--}1300 \text{ mg dl}^{-1}$), 34 mg dl^{-1} IgM (normal $30\text{--}120 \text{ mg dl}^{-1}$), and 184 mg dl^{-1} IgA (normal $20\text{--}230 \text{ mg dl}^{-1}$). He had a normal absolute lymphocyte count and normal percentages of T cells and B cells. Specific antibodies against 15 different pneumococcal polysaccharides or against tetanus toxoid (with which he had also been immunized) were not present in his serum. Because of the severity of his infection, his hypogammaglobulinemia, and lack of specific antibody, intravenous immunoglobulin replacement therapy was begun at a dose of 400 mg kg^{-1} body weight every 3 weeks.

As Robert grew, it became clear that he had dysplastic ectoderm (that is, an abnormality in the growth of the structures produced from the ectoderm). His hair never grew properly; his first teeth erupted at 21 months and were pointed and conical (Fig. 23.3). A skin biopsy revealed a lack of eccrine sweat glands (hypohidrosis). On the basis of these findings, his hypogammaglobulinemia, and his clinical history, the diagnosis of ectodermal dysplasia with immunodeficiency was made. A mutation in the NEMO gene was suspected. DNA sequence analysis revealed a point mutation that had caused an amino acid change at position 417 at the carboxy-terminal end of the NEMO protein.

Robert remained well until he was 2 years old, when he developed a hyperpigmented lacy rash on his back. A biopsy revealed diffuse granulomatous inflammation, and an acid-fast stain was positive for bacilli. A culture of the skin sample grew *Mycobacterium avium*. The cutaneous atypical mycobacterial infection was treated with a combination of antibiotics chosen on the basis of the *in vitro* sensitivity of the isolated mycobacteria. After several months of antibiotic treatment the rash had

Nine-month-old boy with strep meningitis. Had been immunized.

Hypogammaglobulinemia and no anti-strep antibodies.

Ectodermal dysplasia apparent. Query: NEMO deficiency.



Fig. 23.3 Infant with X-linked ectodermal dysplasia and immunodeficiency due to NEMO deficiency. The features of ectodermal dysplasia include frontal bossing, deep-set eyes, fine or sparse hair, and conical or missing teeth.

disappeared, but repeated attempts to discontinue the antibiotics failed, because the rash always returned. Robert is currently well on immunoglobulin replacement therapy and continuous treatment with anti-mycobacterial antibiotics.

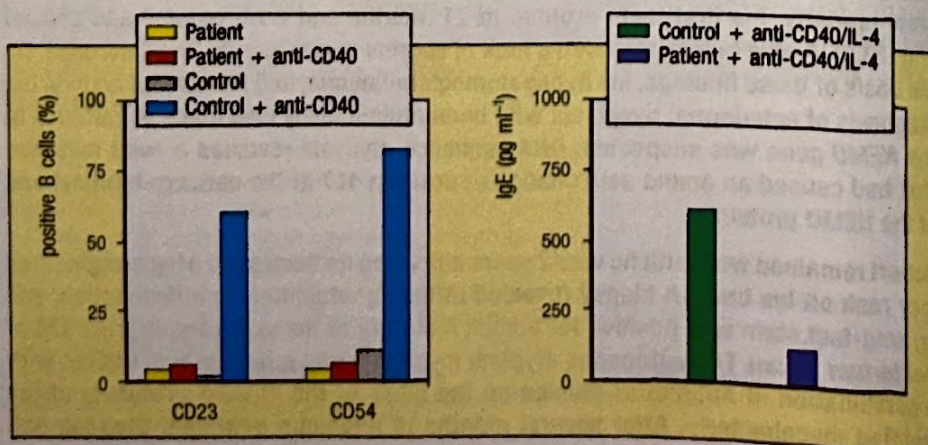
NEMO deficiency and X-linked hypohidrotic ectodermal dysplasia with immunodeficiency.

X-linked ectodermal dysplasia with immunodeficiency is caused by hypomorphic mutations in the *NEMO* gene that lead to the production of a protein with impaired function. This results in immunodeficiency because the end point of the intracellular signaling pathways leading from a variety of receptors important in activating immune responses is activation of NF κ B and the induction of gene transcription. In the absence of any component of IKK (such as NEMO), I κ B is not phosphorylated and so it remains bound to NF κ B, thus preventing the translocation of NF κ B to the nucleus. A complete absence of NEMO function, an amorphic mutation, is lethal in the embryo. The receptors leading to NF κ B activation include the Toll-like receptors on macrophages, which recognize bacterial lipopolysaccharide (LPS), peptidoglycans, and bacterial DNA (CpG), and whose engagement stimulates innate immunity. Other receptors include CD40, which is important in adaptive immunity. In patients with a hypomorphic *NEMO* mutation, the interaction between CD40 on B cells and CD40 ligand on T cells occurs but NF κ B activation is impaired, leading to diminished activation of mature naive B cells. This results in impaired antibody synthesis and impaired switch class recombination (as assessed by a lack of upregulation of the surface markers CD23 and CD54 (ICAM-1) and decreased IgE synthesis; Fig. 23.4).

Patients with mutations in *NEMO* have ectodermal dysplasia because a receptor that is required for ectodermal development also depends on IKK function and NF κ B activation to transmit a signal to the nucleus. The receptor is the ectodysplasin A receptor (a member of the TNF receptor superfamily), which binds ectodysplasin A (a member of the TNF superfamily). Although defects in the genes for these two proteins result in ectodermal dysplasia, there is no concomitant immunodeficiency because the function of the IKK complex is intact. Interestingly, several patients with immune deficiency due to hypomorphic mutations in *NEMO*, but without ectodermal dysplasia, have been identified. Furthermore, distinct hypomorphic *NEMO* mutations have been characterized that differentially impair innate and adaptive immune functions.

The *NEMO* gene is located on the X chromosome, and thus the disease is transmitted by mothers to their sons. The gene contains 10 exons and is 419 amino acids long. Most of the mutations in the *NEMO* gene that cause ectodermal dysplasia

Fig. 23.4 Impairment of NF κ B activation and B-cell activation in a patient with ectodermal dysplasia and immunodeficiency due to a mutation in *NEMO*. Left panel: B-cell activation can be assessed by increases in surface expression of CD23 (the low-affinity IgE receptor) or CD54 (ICAM-1). The presence of these molecules on the surface of B cells is measured by flow cytometry. In a healthy individual, stimulation through CD40 results in the majority of B cells expressing CD23 and CD54 on the cell surface. The gray bars show the unstimulated B cells, and the blue bars show the stimulated B cells. No increased expression of CD23 and CD54 after CD40 ligation is seen in the patient's B cells: compare the unstimulated and stimulated patient's B cells (yellow and red, respectively). Right: CD40 ligation causes IgE isotype switching and IgE synthesis in normal B cells (green) but not in B cells from a NEMO patient (purple), further demonstrating their inability to transmit a signal from CD40.



are located in the tenth exon, which encodes a zinc-finger domain that is a site of protein-protein interaction.

Women who carry a single copy of a *NEMO* gene with a point mutation generally do not have overt symptoms, although they express both normal *NEMO* and mutant *NEMO*. Some may have missing teeth, mildly abnormal patterns of hair or unusual birthmarks. However, a major frameshift or deletion mutation in *NEMO* results in a condition known as incontinentia pigmenti, which is characterized by dermal scarring and hyperpigmentation in heterozygotes. This disorder does not affect the immune system, because the normal *NEMO* gene allows appropriate immune-system development and function. In fact, the immune cells of these women display nonrandom X-chromosome inactivation, and all their immune-system cells contain the normal *NEMO* gene. Any male offspring of women with incontinentia pigmenti who receive the mutant *NEMO* gene die *in utero*.

Questions.

- 1 Why are boys with *NEMO* mutations susceptible to infection with opportunistic mycobacteria such as *M. avium*?
- 2 Patients with *NEMO* mutations also suffer from chronic viral infections such as cytomegalovirus (CMV). Why is this?
- 3 Although B cells from some patients with *NEMO* mutations fail to undergo immunoglobulin class switching in response to CD40 ligation, these patients have residual serum levels of IgG and IgA. Why is this?
- 4 Recently a patient has been discovered to have a gain-of-function mutation in I κ B α . This patient, who also has ectodermal dysplasia, has a more severe immunological defect than patients with the *NEMO* defect. The gain-of-function mutation results from the change of a serine residue to an isoleucine residue, so that the I κ B α chain cannot be phosphorylated. Why do you think the immunodeficiency is more severe in this defect?
- 5 You follow a family in your immunodeficiency clinic. There are two brothers and a maternal male cousin who suffer from recurrent pneumonias. One patient has had an infection with an atypical mycobacterium. None of the boys has ectodermal dysplasia. Could this family still have a hypomorphic *NEMO* mutation? You have sequenced *NEMO* in cDNA generated from one of the boys to rule out a mutation. No mutation was found. Does the lack of a mutation in the patient's *NEMO* coding sequence rule out *NEMO* as a cause of the immune deficiency?